
3.5.8 Gene cloning technologies allow study and alteration of gene function in order to better understand organism function and to design new industrial and medical processes.

Gene cloning and transfer

Fragments of DNA can be produced by

- conversion of mRNA to cDNA, using reverse transcriptase
- cutting DNA at specific, palindromic recognition sequences using restriction endonucleases
- using the polymerase chain reaction (PCR) to form multiple copies of DNA fragments.

Genes can be cloned by *in vivo* and *in vitro* techniques.

In vivo cloning. The use of restriction endonucleases and ligases to insert DNA fragments into vectors, which are then transferred into host cells. The identification and growth of transformed host cells to clone the desired DNA fragments. The importance of “sticky ends”.

In vitro cloning. The use of the polymerase chain reaction (PCR) in cloning DNA fragments.

The relative advantages of *in vivo* and *in vitro* cloning.

The use of recombinant DNA technology to produce transformed organisms that benefit humans.

Candidates should be able to

- interpret information relating to the use of recombinant DNA technology
- evaluate the ethical, moral and social issues associated with the use of recombinant technology in agriculture, in industry and in medicine
- balance the humanitarian aspects of recombinant DNA technology with the opposition from environmentalists and anti-globalisation activists.

Gene therapy

The use of gene therapy to supplement defective genes.

Candidates should be able to evaluate the effectiveness of gene therapy.

Medical diagnosis	The use of labelled DNA probes and DNA hybridisation to locate specific genes. Once located, the base sequence of a gene can be determined by • restriction mapping • DNA sequencing. Many human diseases result from mutated genes or from genes that are useful in one context but not in another, e.g. sickle cell anaemia. DNA sequencing and the PCR are used to produce DNA probes that can be used to screen patients for clinically important genes. The use of this information in genetic counselling, e.g. for parents who are both carriers of defective genes and, in the case of oncogenes, in deciding the best course of treatment for cancers. Candidates should understand the principles of these methods. They should be aware that methods are continuously updated and automated.
Genetic fingerprinting	An organism's genome contains many repetitive, non-coding base sequences. The probability of two individuals having the same repetitive sequences is very low. The technique of genetic fingerprinting in analysing DNA fragments, that have been cloned by PCR, and its use in determining genetic relationships and in determining the genetic variability within a population. Candidates should be able to • explain the biological principles that underpin genetic fingerprinting techniques • interpret data showing the results of gel electrophoresis to separate DNA fragments • explain why scientists might use genetic fingerprints, in the fields of forensic science, medical diagnosis, animal and plant breeding.

Biological principles

When they have completed this unit, candidates will be expected to have an understanding of the following principles.

- Organisms regulate their internal environment and so maintain optimum conditions for their metabolism.
- Animals respond to their internal and external environment as a result of stimulus perception, chemical and electrical coordination and a response by effectors. Plants respond to their external environment as a result of specific growth factors that regulate cell growth.
- The genetic code is held in the base sequence of nucleic acids and determines the amino acid sequence of polypeptides produced by a cell.
- Regulating gene expression enables a cell to control its own activities and development.
- Scientists are able to manipulate gene expression for many agricultural, industrial and medical purposes.

Understanding of these principles, together with those established during the study of other units, will contribute to the assessment of synoptic skills in Unit 5.

Investigative and practical skills

Opportunities to carry out the necessary practical work should be provided in the context of this unit. Candidates will be assessed on their understanding of *How Science Works* in all three A2 units. To meet this requirement, candidates are required to

- use their knowledge and understanding to pose scientific questions and define scientific problems
- carry out investigative activities, including appropriate risk management, in a range of contexts
- analyse and interpret data they have collected to provide evidence
- evaluate their methodology, evidence and data, resolving conflicting evidence.

They will require specific knowledge of the following skills and areas of investigation. The practical work undertaken should include investigations which consider the effect of external stimuli on taxes and kineses in suitable organisms.